

Technical Document #63: Database Systems - Comprehensive Overview

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Database migration tools manage schema changes and data transformations. Flyway and Liquibase are popular database migration tools.

Sharding distributes data across multiple database servers. It improves performance by parallelizing query processing.

Database replication copies data across multiple servers. It improves availability and provides fault tolerance.

Cybersecurity threats continue to evolve in complexity and scale. Organizations must invest in robust security measures to protect sensitive data.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Artificial intelligence continues to reshape industries from healthcare to finance. Machine learning models can now perform tasks that were once thought to require human intelligence.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Key Takeaways:

- Sharding distributes data across multiple database servers.
- Relational databases organize data into tables with rows and columns.
- Database migration tools manage schema changes and data transformations.